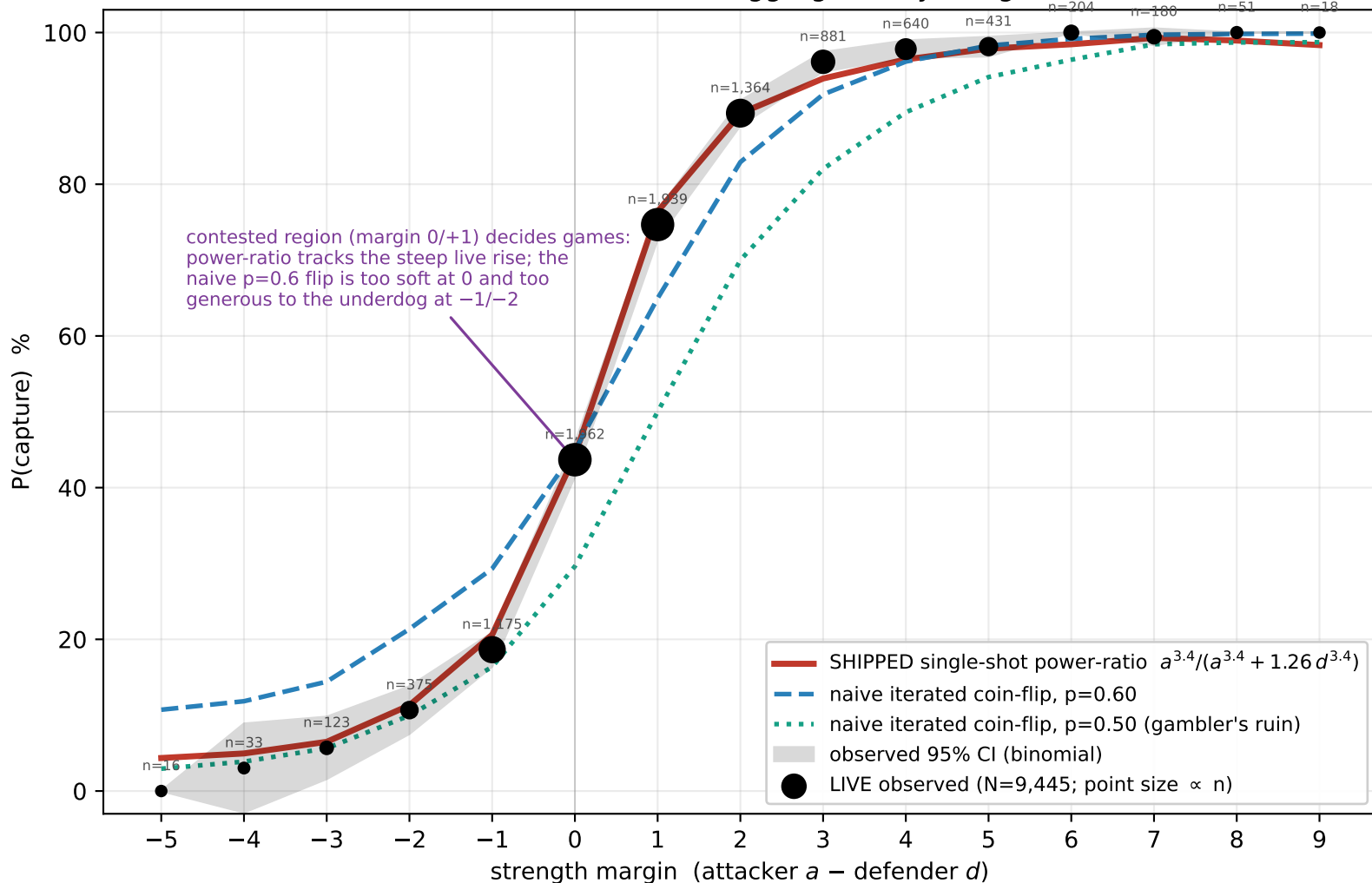
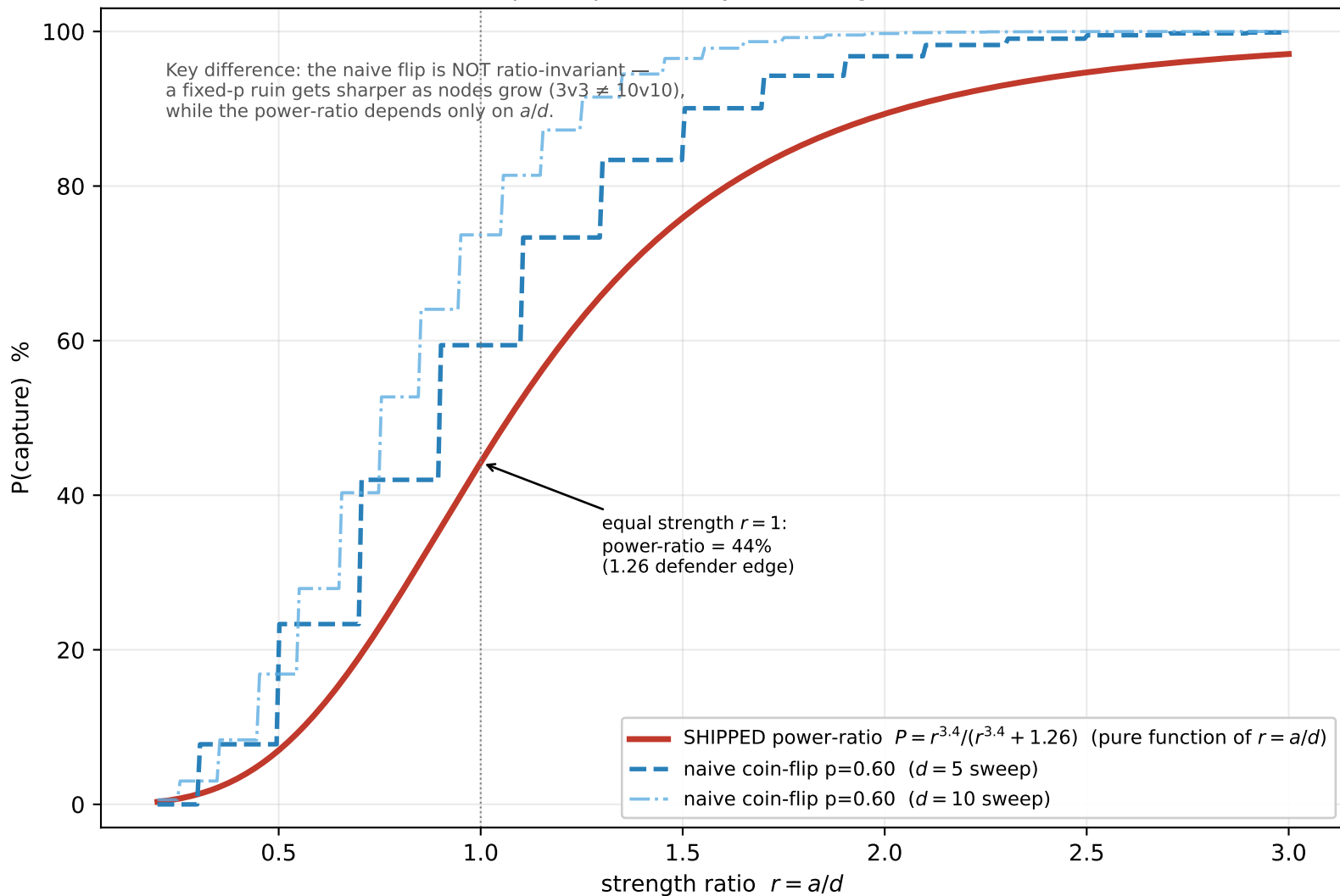


Battle model vs reality (each model scored on the real (a, d) of all 9,445 live battles, aggregated by margin)



Capture probability vs strength ratio

Key difference: the naive flip is NOT ratio-invariant
 a fixed-p ruin gets sharper as nodes grow (3v3 $\neq$ 10v10),
 while the power-ratio depends only on a/d .



Fit to 9,445 live battles — models scored on real (a, d)

margin	n	live %	power-ratio %	err	naive 0.6 %	err
-3	123	5.7	6.5	0.8	14.4	8.7
-2	375	10.7	11.3	0.6	21.4	10.7
-1	1,175	18.6	20.6	2.0	29.3	10.7
+0	1,962	43.7	44.2	0.6	44.9	1.2
+1	1,939	74.7	76.4	1.7	65.0	9.7
+2	1,364	89.4	89.3	0.0	82.9	6.4
+3	881	96.1	93.9	2.2	91.8	4.3
+4	640	97.8	96.5	1.4	96.2	1.6
+5	431	98.1	97.9	0.3	98.3	0.1
+6	204	100.0	98.5	1.5	99.2	0.8
RMSE			1.4		6.9	

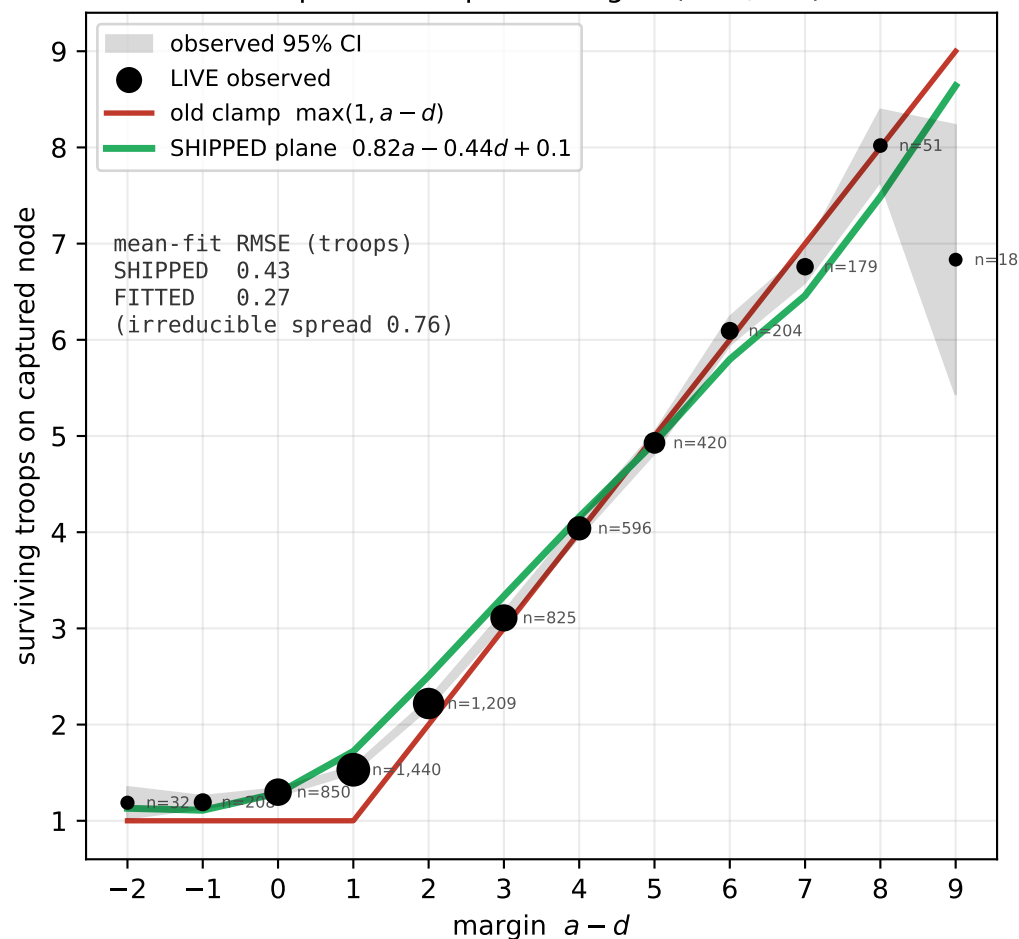
Per-battle goodness of fit (all 9,445 battles)

log-likelihood: power-ratio -3,876 > naive 0.6 -4,053 > naive 0.5 -4,484 (higher better)
 Brier score: power-ratio 0.1324 < naive 0.6 0.1397 < naive 0.5 0.1570 (lower better)

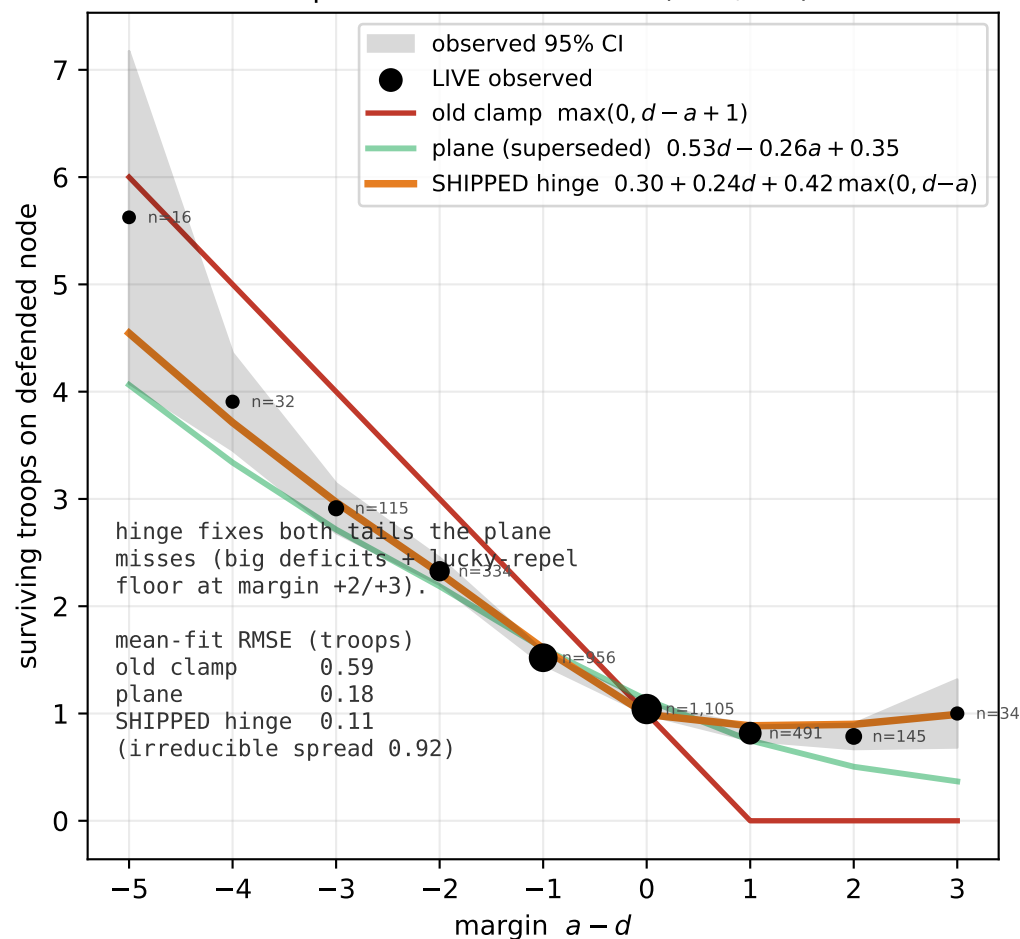
Matches BATTLE_FUNCTION.md §6: single-shot power-ratio is both the simplest model (one closed-form Bernoulli, no loop) and the best fit. The naive fixed-p coin-flip is too soft at the contested margins.

Beyond win/loss: how many troops remain. Source node $\rightarrow 1$ in $\sim 100\%$ of battles. Fitted (a, d) planes beat the margin clamp.

Capture: occupier strength (n=6,081)



Repel: defender remnant (n=3,259)



Dropped 105 physically-impossible survivor reads (occupier $> a - 1$ or remnant $> d - \text{OCR/reinforcement artifacts that inflated the rare-outcome tails}$).